ETHIOPIAN PAY-LATER PLATFORM

Technical Architecture & Development Proposal

Production-oriented technical proposal; no application code

Addis Ababa-first • Provider-independent

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CONFIDENTIAL WORKING DOCUMENT

Document Control

Version	Date	Prepared By	Status	Summary
1.0	21 August 2026	Eleanor Tefera(Fintech)	Working baseline	Initial researched package release

Research Status Legend

VERIFIED	Directly supported by an official/primary source.
REPORTED	Supported by a reputable third-party source.
INFERENCE	Reasonable conclusion from available evidence.
ASSUMPTION	Design assumption for planning; not presented as market fact.
REQUIRES CONFIRMATION	Needs provider, bank, NBE, legal or business-partner confirmation.

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1. Technical Executive Summary

Recommend a modular monolith for the first production platform: one deployable backend with strongly separated business modules, PostgreSQL as the authoritative relational database, Redis for cache/locks/queues, Flutter for customer and merchant applications, and a modern web admin. This minimizes early operational complexity while preserving module boundaries that can later be extracted if scale/team ownership justifies it.

2. Architecture Decision

Option	Benefit	Risk	Recommendation
Modular monolith	Fast delivery, transactional consistency, easier testing/operations.	Requires discipline to preserve module boundaries.	SELECT for MVP/pilot.
Service-based	Independent deployment for selected domains.	Distributed failure/observability complexity.	Use later for clearly justified provider/reporting workloads.
Microservices	Independent scale/team ownership.	High operational and financial consistency complexity.	Do not start here.

3. High-Level Architecture

Customer Flutter App + Merchant Flutter App + Admin Web + Enterprise Merchants → API/Application Layer → Identity/KYC | Customer/Merchant | Credit/Risk | Order/BNPL | Installments | Payments | Ledger/Settlement | Refunds/Disputes | Notifications/Support → PostgreSQL + Redis/Queues + Secure Object Storage → External Credit/KYC/Payment/Notification Partners

4. Technology Stack

Layer	Recommendation	Why	Alternative / When
Backend	Laravel / current supported PHP	Strong productivity, validation/auth/queues/scheduler/ecosystem; suitable for modular monolith and transactional workflows.	Java/.NET/Node can work; change only for team/enterprise constraints, not fashion.
Customer/Merchant Mobile	Flutter	One codebase for Android-first Ethiopia and iOS; strong UI control and offline caching.	Native when device/payment SDK constraints force it.
Admin	React/Next.js or Laravel + React	Rich data-heavy admin and role-based workflows.	Server-rendered Laravel views for smaller internal MVP if speed matters.
Database	PostgreSQL	Strong constraints, transactional behavior, NUMERIC, indexes and JSON where justified.	MySQL acceptable if team standard, but PostgreSQL preferred for financial domain rigor.
Cache/Locks/Queues	Redis + Laravel queues	Rate limits, short-lived cache, distributed locks, worker queues.	Managed queue service at larger scale/ops maturity.
Object Storage	S3-compatible private bucket	KYC/docs/evidence with encryption, lifecycle and signed URLs.	Provider-specific secure object storage.
Notifications	SMS + FCM/APNs + email	OTP and financial reminders/alerts.	Multiple SMS providers for resilience later.
Deployment	Containerized workloads on reputable cloud/managed services	Repeatable deploys, managed DB/backup, observability.	VMs only if operational/regulatory constraints require them.
CI/CD	Git-based pipeline with gated production approval	Automated quality with human approval for sensitive releases.	Equivalent enterprise CI platform.

5. Core Modules

Module	Responsibilities
Identity & Access	Accounts, login, OTP, roles, sessions, devices.

KYC/KYB	Customer/merchant verification, documents, manual review.
Customer & Merchant	Profiles, branches, employees, merchant status.
Credit & Risk	First-time eligibility, limits, reservations, transaction decisions, offers, fraud signals.
Orders & BNPL	Sales, checkout, plan selection, customer acceptance.
Installments	Schedules, due dates, repayment allocation, overdue states.
Payments	Provider-independent attempts, confirmation, retries, webhooks/polling.
Ledger & Settlement	Double-entry postings, merchant payable, settlement batches/items.
Refunds/Disputes	Full/partial refunds, evidence, adjustments.
Reconciliation	Provider/bank/internal matching and exceptions.
Notifications/Support	SMS/push/email, tickets and complaint SLA.
Audit/Configuration	Immutable audits and versioned business/provider settings.

6. Money Representation

Never use binary floating-point for authoritative money. Store ETB amounts as integer minor units (cents/santim equivalent where product policy uses them) or PostgreSQL NUMERIC with a strict rounding policy. Every API and ledger contract must define currency and scale.

7. Concurrency and Idempotency

- Reserve credit exposure atomically before purchase activation.
- Use unique idempotency keys on checkout, payments, refunds and settlements.
- Use database unique constraints for provider references and event IDs.
- Process webhook events through an inbox table; use outbox pattern for reliable event publishing where needed.
- Use row/version locks for concurrent limit reservations and settlement posting.

8. Payment Provider Layer

PaymentService → PaymentProviderInterface → TelebirrAdapter | CbeBirrAdapter | MpesaEthiopiaAdapter | BankAdapter | GatewayAdapter

- Core business modules work with internal payment commands/states, not provider payloads.
- Adapters translate internal requests and provider responses.
- Adapters declare capability flags: initiate, query, refund, mandate, sandbox, webhook, settlement info.
- Unavailable/unconfirmed capabilities remain disabled in production configuration.

9. Customer Mobile Application

- Secure onboarding, KYC and first-time eligibility.
- Home with approved/available amount and next due installment.
- Merchant discovery/QR/checkout and transparent plan disclosure.
- Payment initiation with explicit pending/reconciliation UI.
- History, support, notifications, security/devices.
- Local cache for read-only data; financial/credit actions remain server-authoritative.

10. Merchant Mobile Application

- KYB/onboarding, branches and staff.
- Quick Sale and dynamic QR.

- Transaction state that explicitly separates APPROVED from WAITING_FOR_FIRST_PAYMENT/PENDING.
- Refund requests and settlement reports.
- Restricted employee permissions and activity audit.

11. Admin Web Platform

- Customer/KYC/KYB reviews
- Credit decisions/limits/manual review
- Payment/reconciliation exceptions
- Refund/settlement operations with separation of duties
- Fraud/disputes/support complaints
- Configuration/version publishing
- Audit/reporting/export with role filtering

12. Enterprise Merchant Connection

Expose versioned REST APIs/webhooks for merchant order creation, quote/eligibility, checkout, status, cancellation/refund and reconciliation exports. Enterprise credentials are scoped to merchant/branches; every write uses idempotency keys and signed/authenticated webhooks.

13. Security Architecture

- MFA for privileged staff; strong password hashing and optional customer PIN/biometric local unlock.
- Short-lived access tokens/session rotation; revoke lost devices.
- TLS everywhere; database/object encryption; field-level encryption for especially sensitive attributes.
- Central secrets manager and key rotation.
- Webhook signature verification when available plus nonce/timestamp/replay prevention.
- WAF/rate limits/bot protection; fraud events from anomalous auth/payment behavior.
- Admin break-glass access logged and reviewed.

14. Reliability / Observability

- Structured logs with correlation IDs, no secrets/PII leakage.
- Metrics for API latency, errors, queue depth, provider success/pending rates, reconciliation breaks and settlement failures.
- Distributed tracing for checkout/payment where practical.
- Health/circuit breaker per external provider.
- Dead-letter queue review and replay runbook.

15. Testing Strategy

- Unit tests for money, schedules, eligibility rules and state machines.
- Integration tests with database/Redis/provider simulators.
- Contract tests against provider sandbox/documentation when available.
- E2E tests for customer + merchant + payment + settlement.
- Security/performance/DR testing before production.
- Replay tests for duplicate/delayed callbacks and concurrent purchases.

16. Technical Risks

Risk	Technical Response
Provider capability changes	Adapter/capability flags; contract tests.
Credit model changes	Configurable creditor/funder and policy versions.
Duplicate callbacks	Inbox/idempotency/unique provider refs.
Race on credit limit	Atomic reservations/locking.
Ledger mismatch	Balanced postings + reconciliation.
Low connectivity	Small payloads, resumable UX, server-authoritative states.
Premature scale complexity	Modular monolith; extract services only with measured need.

17. Final Technical Recommendation

Build a small but financially rigorous modular platform. Complexity should be concentrated in correctness—credit decisions, payment state, ledger, reconciliation, security and audit—not in premature microservices.

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https://developer.ethiotelecom.et/docs/GettingStarted
Supports: Publicly documented Telebirr business integration categories including C2B, B2B, H5, Mini App and subscription payment.

[TEL-02] VERIFIED — Ethio telecom. Telebirr H5 C2B Web Payment Integration.
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Supports: Public create-order, checkout, query/status, callback and refund integration documentation.

[TEL-03] VERIFIED — Ethio telecom. Telebirr Schedule Payment / CreateDisburseOrder.
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[CBE-01] VERIFIED — Commercial Bank of Ethiopia. CBE Birr.
https://combanketh.et/ways-of-banking/cbe-birr
Supports: Official description of CBE Birr customer/mobile-money services.

[CBE-02] VERIFIED — Commercial Bank of Ethiopia. CBE Products and Services.
https://combanketh.et/cbeapi/uploads/CBE_Products_and_services_English_84e511c740.pdf
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https://www.safaricom.et/

Supports: M-PESA Ethiopia supports merchant payments/business arrangements; exact BNPL API onboarding and sandbox require provider confirmation.

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Client-provided PDF reviewed 21 August 2026
Supports: Customer/merchant/PayLater roles, eligibility, 25% first payment, PayMore, limits, repayment, default, restructuring, disputes, data handling and account rules.

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https://paylaterapp.com/terms-and-conditions/
Supports: Current public PayLater Qatar contractual product model.

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Supports: Phone OTP, QID capture and liveness-based customer onboarding.

[QTR-PL-03] VERIFIED — PayLater Qatar. What is my credit limit and how much can I spend?.
https://help.paylaterapp.com/en/support/solutions/articles/154000139303-what-is-my-credit-limit-and-how-much-can-i-spend-with-paylater-
Supports: Credit limit uses creditworthiness, PayLater repayment history and account usage.

[QTR-PL-04] VERIFIED — PayLater Qatar. Where can I use PayLater?.
https://help.paylaterapp.com/en/support/solutions/articles/154000139291-where-can-i-use-paylater-
Supports: Use across online and in-store merchants.

[QTR-CBQ-01] VERIFIED — Commercial Bank Qatar / Vodafone Qatar. Vodafone Easy-Buy / CBQ Buy Now Pay Later.
https://www.vodafone.qa/en/buynowpaylater
Supports: Bank-card installment model for online and selected physical stores; full purchase reduces card limit and customer repays equal monthly installments.

[QTR-VF-01] VERIFIED — Vodafone Qatar. Vodafone Partners with Commercial Bank to Offer BNPL.
https://www.vodafone.qa/en/about-us/media/press-release/vodafone-partners-with-commercial-bank-to-offer-buy-now-pay-later-instalment-plan
Supports: Qatar device-installment model through bank partnership and merchant stores.